

Muhammad Moiz Bukhari

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Experience

Software Engineer

May 2026 – Present

TWM Wholesale

Stockton, CA

- Built 10+ automation scripts for inventory, procurement, and supplier workflows, reducing ~3–4 hours/day of manual data entry.
- Improved script reliability by diagnosing bottlenecks and resolving failures in high-volume automation pipelines.

Systems Engineer CO-OP

Jan 2025 – Aug 2025

Abbott Laboratories

Pleasanton, CA

- Designed an Arduino-based power automation fixture with relay control, color sensor feedback, and battery detection logic to automate long-duration testing.
- Enabled 72-hour unattended endurance test cycles across multiple fixtures by combining embedded C/C++ firmware with Python test sequencing and logging.
- Verified BLE communication scenarios for HeartMate Touch Monitor and BLE Adapter through formal V&V procedures and traceable test documentation.
- Authored IQ/OQ packages and change-order documentation in Windchill while supporting FDA-regulated product development workflows.

Teaching Assistant

Jan 2022 – Dec 2025

University of the Pacific

Stockton, CA

- Mentored 30+ students per semester in C/C++, data structures, and algorithms through labs, office hours, and code reviews.

Projects

Power Automation Fixture | Embedded C/C++, Python, Arduino

2025

- Built a microcontroller-based automation system for battery-powered device testing with relay control, sensor feedback, and LCD status prompts.
- Reduced test setup and logging overhead by automating sequencing and data collection across multiple fixture configurations.

End-to-End Encryption Messaging App | Python, Full-Stack, Network Programming

- Built a secure peer-to-peer messaging app with AES-based encryption and custom Diffie-Hellman key exchange.
- Implemented client-server communication with persistent session handling and real-time encrypted message delivery.

WaterCon – Cloud IoT Platform | Python, REST API, SQL

- Developed a cloud-connected IoT platform with REST APIs, normalized SQL storage, and Python ingestion pipelines.
- Delivered real-time dashboards for live sensor telemetry and faster reporting workflows.

Skills

Languages: Python, C, C++, SQL, MATLAB, Bash, PowerShell

Embedded & Hardware: Arduino, Embedded Firmware, UART/SPI/I2C, GPIO, BLE, Relay Control, Sensor Integration, ADCs, Soldering, Schematic Reading

Software & Testing: Data Structures, Algorithms, OOP, V&V, Automated Testing, Root-Cause Analysis, Python Scripting, Test Documentation

Tools: Git, Windchill PLM, LabVIEW, Arduino IDE, MPLab, SolidWorks, VMware, Linux, Microsoft Office

Regulatory: FDA 21 CFR Part 820, ISO 13485, IEC 62304, GDP, IQ/OQ/PQ

Education

University of the Pacific

Stockton, CA

B.S. Computer Science, Honors | GPA: 3.83 | Dean's List (All Semesters)

Aug 2021 – Dec 2025

Certifications

Scrum Master | Scrum Alliance

2023

Embedded Systems Programming | AVR

2024